



Nathan Corral

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As a Computer Engineer with a master's specialization in AI and experience in software engineering, I am seeking a full-time role in development of advanced Machine Learning applications to support the digitization of health care.

Job Experience

- **Humanoid Robots Lab** 09.2021 – 09.2022
Research Assistant Bonn, Germany
 - Programmed the ROS interface for 3D localization of humans from an RGBD camera using deep learning and implemented this on a real robot for autonomous navigation.
 - Used the photo-realistic simulator iGibson (Py-Bullet backend) to generate data for a deep reinforcement learning-based path planning algorithm.
- **Head Rush Technologies** 12.2019 – 04.2020
Contract Engineer Boulder, USA
 - Contracted to code the firmware on a ATmega328PB Microchip for a proof-of-concept system.
 - Completed field tests and project documentation.
- **Aqronos** 11.2018 – 12.2019
Software Engineer Denver, USA
 - Designed ROS nodes for visualization of the company's LiDAR prototype.
 - Interacted with a REST API to set parameters on an embedded system.
 - Filtered point clouds and grouped objects using the C++ Point Cloud Library.
- **Creative Edge LLC** 08.2017 – 09.2018
Software Engineer Denver, USA
 - Developed applications for cryptocurrency mining in both Windows and Linux.
 - Wrote software managing OS drivers, system configurations, and 3rd party tools.

Education

- **Rheinische Friedrich-Wilhelms-Universität Bonn** 10.2020 – 09.2023
M.Sc. *Computer Science* Note: 1.7
Thesis: *Stochastic Transformer for Prediction of Multiple Futures*
 - Developed a novel transformer-based predictor architecture, able to learn a distribution over potential futures.
 - Detailed comparison against other stochastic-based models in video prediction, boasting higher structural similarity in frame-wise comparisons.
- **University of Illinois Urbana-Champaign** 08.2013 – 05.2017
B.Sc. *Computer Engineering* GPA: 3.55/4.0

Projects

🔖 ROS 2 Computer Vision

2024

Author

[Video](#), [Source](#)

- Designed a ROS 2 pipeline to run multiple Computer Vision (CV) tasks in parallel.
- Re-index the model output labels, which may be trained on different datasets, into a universal database using NoSQL.
- Run the pipeline on both live camera feed and a dataset, which allowed time comparisons between the asynchronous running of multiple models.

🔖 Semantic Search using Facebook AI Similarity (FAISS)

2024

Author

[Source](#)

- Implemented the first steps in Retrieval-Augmented Generation (ending before "Generation").
- Programmed web-scraping, dataset embedding, and similarity comparisons to recover matches in the dataset from a natural language query.

Skills

Languages	🔖	• English (Native) • German (fluent, C1 self-assessed)
Strengths	🔖	• Problem Solving • Cross-Team Collaboration • Reliable • Technical Documentation • Hard Working
Coding	🔖	• Python • C++ • Bash • C • LaTeX • Java • Go
Software	🔖	• Linux/Ubuntu • GitHub • Docker • ROS/ROS2 • QEMU • Hyperstack • AWS EC2
Libraries (Py)	🔖	• PyTorch • Hugging Face • TensorFlow • Matplotlib • Pandas • OpenCV • NumPy • Scikit-learn
Knowledge	🔖	• Agile • GoogleTest/Pytest • Test-driven Development • POSIX • Databricks • ML Flow • Continuous Integration (CI) • Object Oriented Programming • Data Structures
Deep Learning	🔖	• Computer Vision • Generative AI • Large Language Models • Gradient Descent Optimization • Retrieval-Augmented Generation • Reinforcement Learning • Point Cloud Processing • CUDA